Midterm 2

Graded

Student

[redacted]

Total Points

18 / 50 pts

Question 1

(no title)

3 / 10 pts

- **+ 10 pts** Correct
- **+ 8 pts** Correct standard matrix, incorrect rank and/or nullity
- **+ 7 pts** Shuffled/transposed standard matrix
- ✔ **+ 3 pts** Progress
- **+ 0 pts** Incorrect
- **− 1 pt** Minor error

Question 2

(no title) **4** / 10 pts

2.1 **(no title)** **3** / 5 pts

+ 5 pts Correct

+ 4 pts Showed that the result holds when we put A in row-echelon form, but did not describe how this translates back to the original matrix

+ 4 pts Mostly correct (identified that the column space is one dimensional, or that rank 1 implies all the columns are in the span of the pivot column) but nontrivial errors or gaps such as

- neglecting that one of the column vectors could be 0
- not explaining what u is
- not explaining why the column vectors are scalar multiples of the desired u

✔ **+ 3 pts** Partially correct but serious gaps in the argument such as

- failing to mention the column space or saying something like "A has one independent vector" or "A is linearly dependent" without explaining what that means
- Only using that the columns of A are linearly dependent, which is not strong enough to give the claim
- vague and not well defined notation
- not explaining why rank 1 gives the desired condition

+ 2 pts Showed some understanding of the problem, such as proving a converse statement or giving an example, but no complete solution due to major gaps

+ 1 pt Did some relevant work but no progress towards a solution

+ 0 pts No progress

2.2 **(no title)** **1** / 5 pts

+ 5 pts Correct

+ 4 pts minor errors or unclear what the "v" is

+ 4 pts Gave a structurally correct proof but errors due to errors in part (a)

+ 3 pts Mostly correct but an incomplete solution due to vagueness or gaps otherwise

+ 2 pts An attempt at a complete solution but with major gaps/errors. Or gave an example or proved a special case

✔ **+ 1 pt** Some relevant work but no solution

+ 0 pts No relevant work

− 1 pt Errors carrying over from part (a)

+ 0 pts Click here to replace this description.

Question 3

(no title) **1** / 10 pts

3.1 **(no title)** **1** / 5 pts

+ 5 pts Correct

+ 2 pts Correct Eigenvalues

✔ **+ 1 pt** Some progress

+ 0 pts Incorrect

3.2 **(no title)** **0** / 5 pts

+ 5 pts Correct

+ 3 pts Imprecise arguments

+ 2 pts no justification/incorrect proof

✔ **+ 0 pts** Incorrect

Question 4

(no title) **8** / 10 pts

+ 10 pts Correct

✔ **+ 8 pts** Minor mistakes / missing or wrong error calculation

+ 6 pts Substantial mistakes

+ 4 pts Major mistake in approach/ lots of missing or incorrect steps

+ 2 pts small progress/no clear steps shown

+ 0 pts No attempt is made/No progress

Question 5

(no title) **2** / 10 pts

5.1 **(no title)** **1** / 5 pts

- **+ 5 pts** Correct
- **+ 4.5 pts** almost correct but there is a small issue
- **+ 2.5 pts** some work done towards solving the problem
- ✔ **+ 1 pt** attempt made
- **+ 0 pts** Left blank or no significant work shown

5.2 **(no title)** **1** / 5 pts

- **+ 5 pts** Correct
- **+ 4.5 pts** almost correct but there is a small issue
- **+ 2.5 pts** some work done towards solving the problem
- ✔ **+ 1 pt** attempt made
- **+ 0 pts** Left blank or no significant work shown

MIDTERM 2
MATH 54

Your name

- [redacted]
-

Student ID number

- [redacted]
-

Instructions: Be sure to write your name and your id number. Write your solutions in the space below the problem. If you run out of space use the back of the page. **Show your work and justify your answer. Indicate final answers by circling them.** Good luck!

1. Let $\mathbb{P}_3$ be the vector space of all polynomials $p(t)$ of degree at most 3. Define a linear transformation $T : \mathbb{P}_3 \to \mathbb{P}_3$ by

$$T(p) = p'' + tp'.$$

Find the matrix of T relative to the basis $\mathcal{B} = \{1, t, t^2, t^3\}$, and find its rank and nullity.

matrix $T = [T(b_1)_B | \cdots | T(b_n)_B]$

We have the transformation defined as:

$[T(1)_B \mid T(t)_B \mid T(t^2)_B \mid T(t^3)_B]$.

conjecture

$$\begin{matrix} 1(p''+tp') & & p''+tp' \\ t(p''+tp') & = & tp''+t^2p' \\ t^2(p''+tp') & & t^2p''+t^3p' \\ t^3(p''+tp') & & t^3p''+t^4p' \end{matrix}$$

$\Rightarrow$ 4×2 matrix and it is $\in \mathcal{B}$.
At most we can have rank $= 3$ by our degree constraint, meaning that $\exists$ one free variable

$3 \times 4 = (3 \times n)(4 \times m)^T = (3 \times n)(m \cdot 4)$

$= (3 \cdot 4)$

rank 1

2. Let A be a 3×4 matrix of rank 1.

(a) Show that there exists a nonzero vector $\mathbf{u}$ in $\mathbb{R}^3$ such that each column of A is a scalar multiple of $\mathbf{u}$; that is, there exist real numbers c_i such that

$$\mathbf{a}_i = c_i\mathbf{u}, \quad i = 1, 2, 3, 4.$$

Direct Proof

We know that $\exists$ at most one pivot, this implies that all other col(A) $\vee$ row(A) are dependent on each other giving null(A) = 3. Thus:

$a_i = c_i u$, with $i = (1,2,3,4)$ where i must have the condition that all of its $c_1, \ldots, c_p$ are $\neq 0$, because if they were 0 this means its linearly independent $\Rightarrow$ we found our rank n candidate that all c_i's depend on

(b) Deduce from part (a) that

$$A = \mathbf{u}\mathbf{v}^T$$

for some nonzero $\mathbf{v}$ in $\mathbb{R}^4$.

Proof by contradiction

$\neg P \Rightarrow \neg R \Rightarrow \ldots \Leftarrow R$

$\neg \exists$ uv^T s.t. $A = uv^T$. We know that A is $\in \mathbb{R}^{3\times 4}$ dimension. $\neg P \Rightarrow$ that for all matrix multiplication, with $(n, m \in \mathbb{R})$ that $(3\times n)(4\times m)^T \neq A$. But this is not true, because

$$(3\times n)(4\times m)^T = (3\times n)(m\times 4) = (3\times 4) \in \mathbb{R}$$

which gives our matrix A $\#$

$$\begin{array}{r} {}^{2}\\ 25 \\ 5 \\ \hline 125 \end{array}$$

$$\begin{array}{r} x\,40 \\ 3\,\overline{)\,125} \\ 120 \\ \hline 5 \end{array}$$

40 r 5

$-16/3 - 9/3 = -25/3$

eigenvalues = det (A − λI)
eigenvectors = Null (A − λI)

3. Consider the matrix
$$A = \begin{bmatrix} 5 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & -3 \end{bmatrix} \sim \begin{pmatrix} 5 & 1 & 0 \\ 0 & 1 & 4/3 \\ 0 & 4 & -3 \end{pmatrix} + R2(-4) \sim \begin{pmatrix} 5 & 1 & 0 \\ 0 & 1 & 4/3 \\ 0 & 0 & 25/3 \end{pmatrix}$$

(a) Find the eigenvalues and eigenvectors of A.

det $(5 \cdot 1 \cdot 25/3)$

$$\begin{pmatrix} 5 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & -3 \end{pmatrix} \to \begin{pmatrix} 5-\lambda & 1 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 25/3-\lambda \end{pmatrix} \vee \begin{pmatrix} 5-\lambda & 1 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 4 & -3-\lambda \end{pmatrix}$$

$5-\lambda=0 \to \lambda=5$ $\vee$ $5-\lambda=0 \to \lambda=5$

$1-\lambda=0 \to \lambda=1$ $\quad 3-\lambda=0 \to \lambda=3$

$25/3-\lambda=0 \to \lambda=25/3$ $\quad -3-\lambda=0 \to \lambda=-3$

(b) Determine whether A is diagonalizable. If A is diagonalizable, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

By E.R.O. we can see that A is diagonalizable.

This implies $\exists$ A^{-1}, by cofactor expansion we can find its inverse

$$\begin{pmatrix} 5 & 1 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & -3 \end{pmatrix} \to \begin{matrix} c_{11} = \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} & C_{12} = \begin{pmatrix} 0 & 4 \\ 0 & -3 \end{pmatrix} & C_{13} = \begin{pmatrix} 0 & 3 \\ 0 & 4 \end{pmatrix} \\ c_{21} = \begin{pmatrix} 1 & 0 \\ 4 & -3 \end{pmatrix} & C_{22} = \begin{pmatrix} 5 & 0 \\ 0 & -3 \end{pmatrix} & C_{23} = \begin{pmatrix} 5 & 1 \\ 0 & 4 \end{pmatrix} \\ C_{31} = \begin{pmatrix} 1 & 0 \\ 3 & 4 \end{pmatrix} & C_{32} = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} & C_{33} = \begin{pmatrix} 5 & 1 \\ 0 & 3 \end{pmatrix} \end{matrix} = \begin{pmatrix} -25 & 0 & 0 \\ -3 & -15 & 20 \\ 4 & 20 & 15 \end{pmatrix}$$

$-9-16=-25$; $=0$; $=0$; $=-3$; $=-15$; $=20$; $=4$; $=20$; $=15$

$$\to \begin{pmatrix} -25 & -3 & 4 \\ 0 & -15 & 20 \\ 0 & 20 & 15 \end{pmatrix} \to \frac{1}{\det(A)} \begin{pmatrix} -25 & -3 & 4 \\ 0 & -15 & 20 \\ 0 & 20 & 15 \end{pmatrix} = A^{-1} = P$$

$125/3$

4. Find a least-squares solution to the following system, and find the least-squares error.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

$x - \hat{x} = \|d\|$
= Least squares error

Def/ $[A^TA \mid A^Tb] \rightarrow$ find its inverse $\vee$ rref to find $\hat{x}$

$(2\times4)(4\times2)$ $\quad(2\times2)$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 6 & 12 \end{pmatrix}$$

$c_{11} = 1+1+1+1 = 4$
$c_{12} = 1+1+1+3 = 6$
$c_{21} = 6$
$c_{22} = 3+9 = 12$

$(2\times4)(4\times1)$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 1 \\ 1 \\ 2 \end{pmatrix} = \quad 2\times1 \quad \begin{matrix} c_{11} = 2+1+1+2 \\ c_{21} = 2+1+1+6 \end{matrix} = \begin{pmatrix} 6 \\ 10 \end{pmatrix}$$

$A = \frac{1}{ad-bc}$ ATA inverse

$$\left(\begin{array}{cc|c} 4 & 6 & 6 \\ 6 & 12 & 10 \end{array}\right) \longrightarrow \frac{1}{(4)(12)-(6)(6)}\begin{pmatrix} 12 & -6 \\ -6 & 4 \end{pmatrix} = \frac{1}{12}\begin{pmatrix} 12 & -6 \\ -6 & 4 \end{pmatrix} =$$

$48 - 36 = 12$

$(2\times2)\quad(2\times1) = 2\times1 = \hat{x}$

$$\begin{pmatrix} 12/12 & -6/12 \\ -6/12 & 4/12 \end{pmatrix} = \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1/3 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 10 \end{pmatrix} = \begin{matrix} c_{11} = 6 + -5 \\ c_{12} = -3 + 10/3 \end{matrix} = \begin{pmatrix} 1 \\ 1/3 \end{pmatrix} = \hat{x}$$

$-4/3 + 10/3 = 1/3$

$$\begin{pmatrix} 2 \\ 1 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1/3 \end{pmatrix}$$

would naturally be next step to find error but the dimensions dont match, implying I did something wrong

$n \times n$

$\begin{pmatrix} \circ & \circ \\ \circ & \circ \end{pmatrix} = \begin{pmatrix} - \cdot & - \cdot \\ - \cdot & - \cdot \end{pmatrix}$ $-A$

$\forall x, y \in \mathbb{R}^n$

$(Ax \cdot y = -x \cdot Ay)$

5. Let A be an $n \times n$ matrix such that $A^T = -A$.
(a) Show that for all $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$

$$A\mathbf{x} \cdot \mathbf{y} = -\mathbf{x} \cdot A\mathbf{y}.$$

Let Ax be made up of $c_1v_1 + \dots + c_pv_p$. Then if $\vec{y} \in \mathbb{R}$ is closed under additivity $\wedge$ homogeneity it follows that Axy $\exists$ $\wedge$ is consistent. In other words:

$y_1(c_1v_1x_1) + \dots + y_p(c_pv_px_p)$

and it follows that if $-x$ is simply x negated, then $\exists$ a case where the multiplication is communitive for

$Axy = -Ayx$ #

(b) Deduce that if λ is a real eigenvalue of A, then $\lambda = 0$.

$\text{Eigenspace}_\lambda = \det(A - \lambda I) = 0$ has the conditions that it is $\neg LI$

$\bar{\lambda} = a - bi \vee \lambda = a + bi$ still holds for

$\lambda = 0$